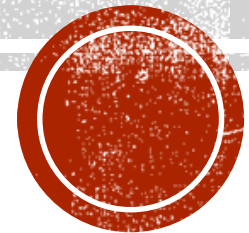


EEE141L: ELECTRICAL CIRCUITS LAB

LAB: 05

EXPERIMENT NAME: DELTA-WYE CONVERSION

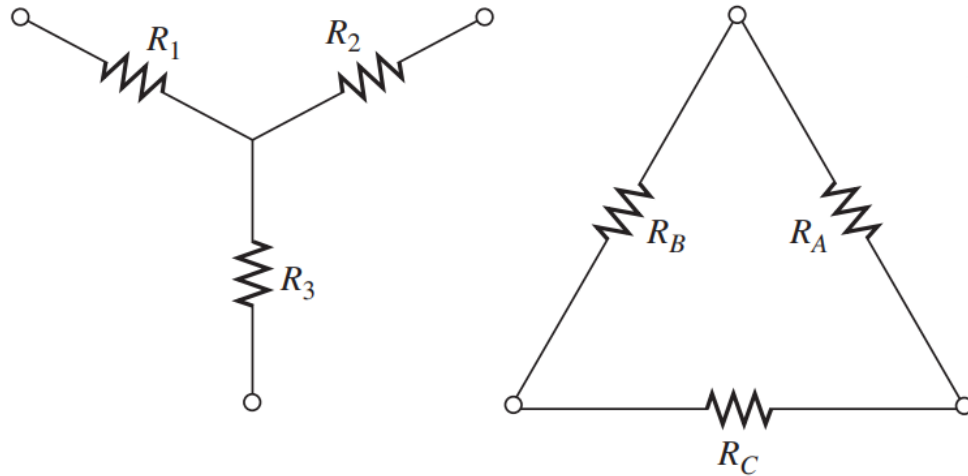
LAB INSTRUCTOR: SUDIPTO SARKER SUPTO



DELTA AND WYE (OR PI AND TEE) CONFIGURATIONS

- Situations often arise in circuit analysis when the resistors are neither in parallel nor in series.
- Under these conditions, it may be necessary to convert the circuit from one form to another to solve for any unknown quantities if mesh or nodal analysis is not applied.
- **Two circuit configurations that often account for these difficulties are the wye (Y) and delta (Δ) configurations.**
- They are also referred to as the tee (T) and pi (π), respectively.

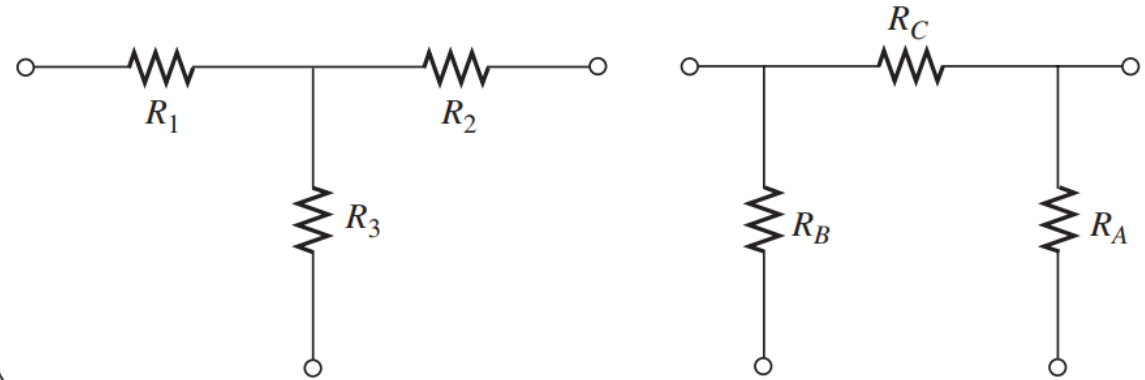
DELTA & WYE (OR PI & TEE) CONFIGURATIONS



“Y”

“ Δ ”

(a)



“T”

“ Π ”

(b)

DELTA TO WYE CONVERSION

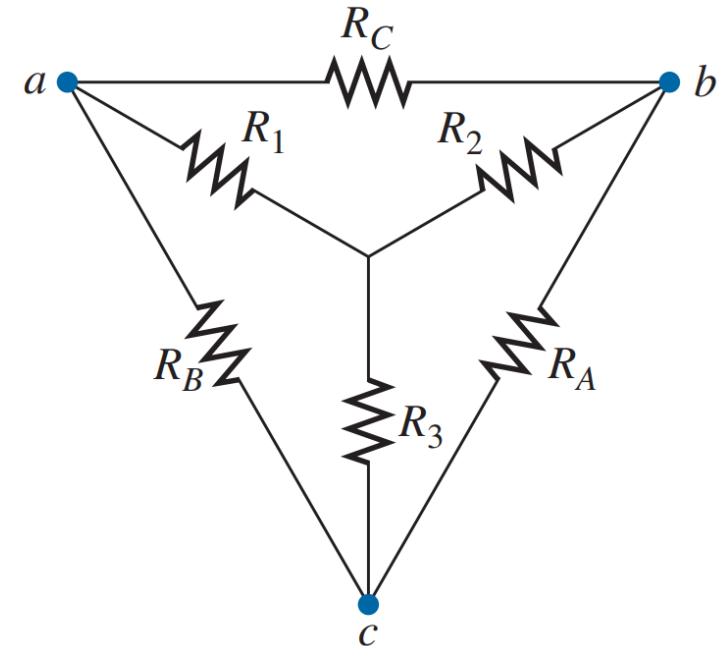
- If a circuit contains a delta configuration, it may be more convenient to convert it into a wye network.
- We superimpose a wye network on the existing delta network and find the equivalent resistances in the wye network.

$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_A R_B}{R_A + R_B + R_C}$$

Note that each resistor of the Y is equal to **the product of the resistors in the two closest branches of the Δ divided by the sum of the resistors in the Δ .**



WYE TO DELTA CONVERSION

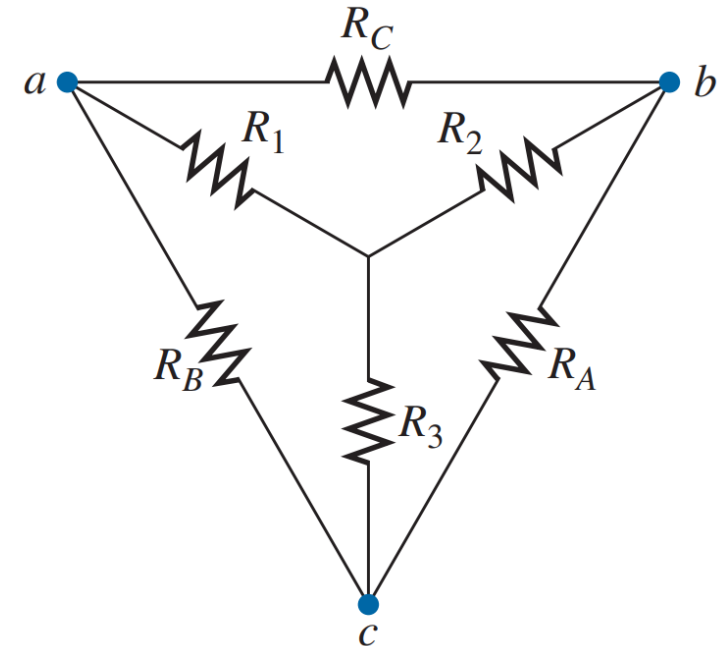
- If a circuit contains a wye configuration, it may be more convenient to convert it into a delta network.
- We superimpose a delta network on the existing wye network and find the equivalent resistances in the delta network.

$$R_A = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$$

$$R_B = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$$

$$R_C = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$$

Note that the value of each resistor of the Δ is equal to **the sum of all possible products of Y resistors taken two at a time, divided by the opposite Y resistor.**



BALANCED DELTA AND WYE NETWORKS

- The Y and networks are said to be balanced when,

$$R_1 = R_2 = R_3 = R_Y$$

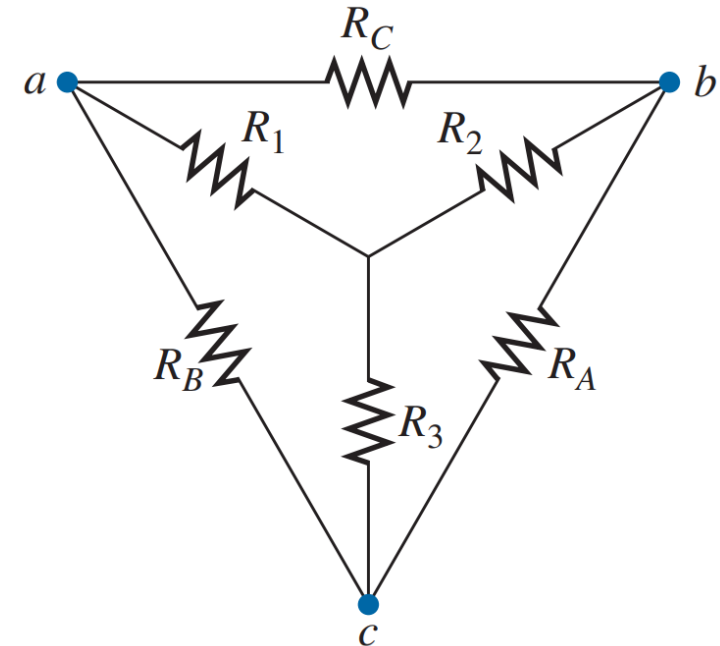
$$R_A = R_B = R_C = R_\Delta$$

- Under these conditions, conversion formulas become,

$$R_Y = \frac{R_\Delta}{3}$$

$$R_\Delta = 3R_Y$$

- This indicates that for a Y of three equal resistors, the value of each resistor of the Δ is equal to three times the value of any resistor of the Y.



THE END...

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